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22 October 1977
SUPERSEDING
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MILITARY SPECIFICATION

ANCHOR, PROPELLANT EMBEDDED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an operational, electrically fired propellant embedded anchor (vice the anchor) having a nominal capacity of 50,000 pounds.

1.2 Classification. The anchor shall be provided in the following configurational types as specified (see 6.2):

Type I (Model 50) - Propellant Embedded Anchor (actuated by command).
Type II (W/Probe) - Propellant Embedded Anchor (actuated automatically by a bottom-sensing probe).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

FF-N-105 - Nails, Brads, Staples, and Spikes: Wire, Cut, and Wrought.

FSC 3835

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FF-S-85	- Screws, Cap, Slotted and Hexagon Head.
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-T-60	- Tape: Packaging, Waterproof.

MILITARY

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-P-323	- Propellant, M2 and M5.
MIL-P-514	- Plates, Identification, Instruction and Marking Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-R-11468	- Radiographic Inspection; Soundness Requirements for Arc and Gas Welds in Steel.
MIL-C-12000	- Cable, Cord, and Wire, Electric Packaging of.

STANDARDS

FEDERAL

FED. STD. No. 356	- Commercial Packaging of Supplies and Equipment.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MS15795	- Washer, Flat - Metal, Round, General Purpose (In./MM).

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MS35338	- Washer, Lock-Spring, Helical, Regular (Medium) Series (In./MM).
MS35494	- Screw, Wood, Flat Head, Slotted, Steel and Brass.
MS51967	- Nut, Plain, Hexagon-Carbon Steel, Cadmium Plated, UNC-2B (In./MM).

DRAWINGS

ME

TA13213E8600	- Anchor Assembly, Propellant Embedded, Type I.
TA13218E8600	- Anchor Assembly, Propellant Embedded, Type II.
TA13218E8601	- Retrofit Kit, Propellant Embedded Anchor, Type II.
TA13201E8933	- Anchor Test and Capacitor Charging Kit.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN WELDING SOCIETY, INC. (AWS)

Codes for Welding

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. 7th Street, Miami, FL 33125.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Welding Qualifications of the American Society of Mechanical Engineers.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

CODE OF FEDERAL REGULATIONS

49 CFR 1-199 - Transportation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Description. The anchors, retrofit kit, and anchor test and capacitor charging kit shall be as shown on the following Top Assemblies and as specified herein:

Type I anchor	- TA13213E8600
Type II anchor	- TA13218E8600
Type II retrofit kit	- TA13218E8601
Type II anchor test and capacitor charging kit	- TA13201E8933

When specified (see 6.2), the retrofit kit and anchor test and capacitor charging kit may be procured separately.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissable without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

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3.2 First article (preproduction model). The contractor shall furnish an anchor, retrofit kit, and anchor test and capacitor charging kit for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods will produce anchors, retrofit kits, and anchor test and capacitor charging kits that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examinations and tests as specified (see 6.2).

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

3.4 Weld filler metals. Steel filler metal for mild steel, corrosion-resistant steel, and special high-strength steel, Grade I shall be in accordance with an applicable code published by the American Welding Society.

3.5 Identification marking. The anchor shall be identified in accordance with MIL-STD-130. The marking shall be applied to the anchor on plates conforming to MIL-P-514, Type I, Style 3, Composition C of Type I, Grade A, Class 1 material. The plates shall be attached by screws, bolts, or rivets in a location where the plates will be visible and legible.

3.6 Treatment and painting. The portions of the anchor normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type B.

3.7 Items to be furnished with the anchors.

3.7.1 Cartridge parts (Type I anchor). All cartridge parts for the Type I anchor, except propellant material (see 3.7.2), shall be supplied unassembled. Sufficient cementing material, as specified, for assembly of the cartridge shall be packed with each anchor.

3.7.2 Propellant material. When specified (see 6.2), propellant, blasting caps, and detonating fuses shall be furnished.

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3.7.3 Cartridges and fuse assemblies (Type II anchor). Unless otherwise specified (see 6.2), one cartridge and one fuse assembly shall be supplied for each anchor, and one spare cartridge and one spare fuse assembly shall be supplied for each five anchors. If the contract or order is for a quantity of anchors which is not a multiple of five, one spare cartridge and fuse assembly shall be supplied for the remaining one, two, three, or four anchors.

3.7.4 Retrofit kit and anchor test and capacitor charging kit (Type II anchor). When specified (see 6.2), the retrofit kit and the anchor test and capacitor charging kit shall be furnished with the Type II anchor.

3.8 Workmanship. All parts, components, and assemblies of the anchor including castings, forgings, molded parts, stampings, and welded parts shall be clean and free from sand, dirt, fins, sprues, scale, flux, and other harmful extraneous material. All edges shall be rounded or chamfered.

3.8.1 Steel fabrication. Steel used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.8.2 Welders and welding.

3.8.2.1 Welders. Before assigning any welder to manual or machine welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society.
Welding Qualifications of the ASME.

3.8.2.2 Steel welding. Steel welding shall be performed by the shielded arc method in accordance with Standard Qualification Procedures of the American Welding Society. Where enbrittlement or harmful residual stresses of the joined metal may result, preheating or annealing of both shall be

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employed. All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. All welding on special high-strength steel shall be in accordance with referenced drawings and shall be checked for soundness in accordance with MIL-R-11468, Standard III. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the anchor to conform to the examination and tests requirements specified in Section 4.

3.8.2.3 Rewelding defects. Portions of welded joints that have been rejected on inspection because of defects may be repaired only by rewelding. The defective area shall be removed by chipping or machining. Before rewelding, the joint shall be inspected to assure that all of the defective weld has been removed and that the joint is accessible so that the welder can obtain the specified weld and quality of the joint.

3.9 Assembly (Type II anchor). Each Type II anchor shall be completely assembled ready for use, except for the cartridge and the probe which shall be packed separately (see 5.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.2 Disassembly inspection. Failure of any test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction anchor shall be examined as specified in 4.5.1. Special attention shall be given the radiographic inspection of all welding on special high-strength steel. This inspection shall conform to MIL-R-11468, Standard III. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction anchor shall be tested as specified in 4.5.2. Failure of this test shall be cause for rejection or performing the inspection specified in 4.1.2.

4.3.2.1 Fuse. The preproduction fuse assembly, before application of the encapsalent shall be tested in accordance with 4.5.2.3. Any fuse assembly which fails this test shall be repaired and retested. After application and curing of the encapsalent, and assembly of the fuse with the anchor; the fuse shall be retested in accordance with 4.5.2.3. Failure of this final test shall result in rejection of the unit pending removal of the encapsalent, repair of the unit, and retesting.

4.3.2.2 Assembled Type II anchor. Each assembled Type II anchor, less cartridge, shall be vacuum tested in accordance with 4.5.2.2. Failure of this test shall be cause for rejection.

4.3.2.3 Cartridge. The preproduction cartridge shall be tested in accordance with 4.5.2.3. Failure to pass the test will result in rejection of the unit.

4.4 Quality conformance inspection.

4.4.1 Examination. Each of the following shall be examined as specified in 4.5.1: Anchor, retrofit kit, fuse assembly, cartridge, and anchor test and capacitor charging kit. Presence of one or more defects shall be cause for rejection.

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4.4.2 Tests. During production, one Type I anchor from each lot (a lot shall consist of not more than 10 anchors) shall be tested as specified in 4.5.2. Failure of this test shall be cause for rejection of the lot.

4.4.2.1 Each of the following shall be tested as specified in 4.5.2: Type II anchor, fuse assembly, cartridge, and anchor test and capacitor charging kit. Presence of one or more defects shall be cause for rejection.

4.5 Inspection schedule.

4.5.1 Examination. The anchor shall be examined for the following defects:

101. Materials not as specified.
102. Dimensions not as shown on drawing. (Special attention shall be paid to I.D. of barrel and O.D. of "O"-ring groove on piston).
103. Components missing or not as specified.
104. Treatment or painting not as specified.
105. Identification marking improper, illegible, or missing.
106. Workmanship not as specified.
107. Failure of welding on special high-strength steel or failure to pass soundness check in accordance with MIL-R-11468 (see 3.8.2.3).
108. Improper assembly (anchor only).

4.5.1.2 Fuse. The fuse shall be examined for the following defects:

109. Materials not as specified.
110. Dimensions not as shown on the drawings.
111. Components missing or not as specified.
112. Identification marking improper, illegible, or missing.
113. Workmanship not as specified.

4.5.1.3 Cartridge. The cartridge shall be examined for the following defects:

114. Materials not as specified.
115. Dimensions not as shown on the drawings.
116. Components missing or not as specified.
117. Propellant quantity not as specified.
118. Workmanship not as specified.

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4.5.1.4 Retrofit kit. The retrofit kit shall be examined for the following defects:

119. Components missing or not as specified.
120. Defective workmanship.
121. Defective components.

4.5.1.5 Anchor test and capacitor charging kit. The anchor test and capacitor charging kit shall be examined for the following defects:

122. Components missing or not as specified.
123. Defective workmanship.
124. Defective components.

4.5.2 Tests.

4.5.2.1 Type I anchor. The anchor barrel and piston (less cartridge) shall be inspected and vacuum tested to 20 inches of mercury. The test line shall connect to the test opening in the lead plug. The vacuum shall be held for 30 minutes. Any evidence of leakage shall constitute failure of this test.

4.5.2.2 Type II anchor. The Type II anchor shall be assembled (less cartridge), with the firing wire through the breech block, and vacuum tested to 20 inches of mercury. The test line shall connect to the test opening in the breech plug. The vacuum shall be held for 30 minutes. Any evidence of leakage shall constitute failure of this test. After testing, sealant shall be applied to the threaded plug and the plug screwed tightly into the test opening. Any disassembly of the anchor invalidates this test, and the test must be repeated upon reassembly of the anchor.

4.5.2.3 Fuse assembly. Each fuse assembly shall be tested as follows:

(a) Connect a voltmeter across the firing wire. Connect the plug between the power supply box and switch box. Install a 22-1/2-volt battery and a 2-amp fuse in the power supply box.
(b) The voltmeter should show no voltage. Slowly pressurize the pressure switch while observing the fuse. The fuse should burn out at about 7-psi pressure. The voltmeter should still show no voltage.

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(c) Keep the pressure switch pressurized and depress the contact switch. The voltmeter should show about 15 volts, and the voltage should slowly decrease.
(d) Keep the contact switch depressed and release the pressure switch. The voltage should rapidly diminish to near zero.

Failure of the 2-amp fuse to burn out at about 7-psi pressure, or voltage readings not as indicated above shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Preproduction pack inspection.

4.6.1.1 Examination. The preproduction packs, of the Type I, and Type II anchors, the Type II retrofit kit, and the test and capacitor charging kit, shall be examined for the defects specified in 4.6.2.2. Presence of one or more defects shall be cause for rejection.

4.6.1.2 Tests. The preproduction pack of the Type II retrofit, kit shall be subjected to the railroad-car test as specified for guided-impact tests in accordance with MIL-STD-1186, Appendix A. The car shall strike a string of five empty cars, with draft gear extended and the brakes set, at a speed of not less than 10 miles per hour. At the conclusion of the preproduction pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking, or bracing, or any visible damage to the contents shall constitute failure of the test and shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.2 Examination. Samples selected in accordance with 4.6.2.1 shall be examined for the following defects. AQL shall be 2.5 percent defective.

125. Materials, methods, and containers not as specified. Each incorrect material, method, or container shall constitute one defect.
126. Unprotected metal surfaces not preserved as specified for Level A.
127. Barrel not preserved as specified for Level A.

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128. Piston plug and piston tube not preserved as specified for Level A.
129. Electrical wire not preserved in accordance with referenced document for Level A.
130. Breech block not preserved and assembled as specified for Level A.
131. Switch box for the retrofit kit not disassembled and preserved as specified for Level A.
132. Power supply assembly cable for retrofit kit not secured as specified for Level A.
133. Openings not sealed as specified for Level A.
134. Pendant for retrofit kit not coiled and secured as specified for Level A.
135. Consolidation not as specified for Level A.
136. Fiberboard container not closed as specified for Level A.
137. Marking illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination and test within the time frame specified (see 6.2) to prove, prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The preproduction pack may be accomplished utilizing either the preproduction model anchor or a production anchor. If the preproduction model anchor is utilized, any preservation, and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the preproduction model anchor and production anchors.

5.2 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

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5.2.1.2 Unprotected surfaces. Unpainted or otherwise unprotected exterior metal surfaces or components such as bolts, nuts, pins, exposed threads, or any other unprotected exterior metal surfaces not specified herein requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative.

5.2.1.3 Type I and Type II anchors.

5.2.1.3.1 Barrel. All interior surfaces of the barrel shall be coated with Type P-11 preservatives.

5.2.1.3.2 Piston plug and piston tube. All surfaces of the piston plug and piston tube shall be coated with Type P-11 preservative prior to final assembly.

5.2.1.3.3 Electrical wire. The electrical wire shall be coiled and placed in a fiberboard box in accordance with the requirements of Level A of MIL-C-12000.

5.2.1.3.4 Breech block assembly. The breech block surface facing inside the barrel, the cable opening in the water seal plug, and the breech block threads shall be completely coated with Type P-11 preservative. The preserved breech block assembly shall then be assembled into the mating end of the barrel. The cable opening through the water seal plug shall be sealed with tape conforming to PPP-T-60, Type IV.

5.2.1.3.5 Cartridge. The unassembled parts of the cartridge (see 3.7.1) shall be preserved in accordance with MIL-P-116, Method IC-1. Like materials may be preserved together except the cartridge case shall be individually preserved, Method IC-1.

5.2.1.3.6 Probe Type II anchor. The probe and probe spring shall be disassembled from the switch box. The probe foot shall be disassembled from the shaft. The probe spring shall be coated with Type P-11 preservative and preserved Method IC-1 of MIL-P-116. The screw shall be reinserted into the shaft and the cotter key preserved, Method IC-1 of MIL-P-116.

5.2.1.3.7 Consolidation. The preserved cartridge and probe parts shall be placed in a close-fitting fiberboard box conforming to PPP-B-636, W5c. The contents shall be cushioned to prevent free movement within the box. The box shall be closed in accordance with Method V of the appendix to the box specification.

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5.2.1.4 Type II retrofit kit (see 3.7.4).

5.2.1.4.1 Piston tube. The piston tube, preserved as specified in 5.2.1.3.2 shall be wrapped in barrier material conforming to MIL-B-121, Type I, Grade A, with the wrap secured with tape PPP-T-60, Type IV. The piston tube shall then be placed in a box conforming to PPP-B-621. The contents shall be blocked and braced to prevent movement and damage.

5.2.1.4.2 Piston plug. The piston plug, preserved as specified in 5.2.1.3.2 shall be wrapped with barrier material conforming to MIL-B-121, Type I, Grade A with the wrap secured with Tape PPP-T-60, Type IV.

5.2.1.4.3 Projectile anchor. The mating surface for the piston plug shall be coated with Type P-11 preservative and sealed with tape PPP-T-60, Type IV. All other surfaces shall be preserved in accordance with 5.2.1. The wire rope shall be coiled to the minimum diameter.

5.2.1.4.4 Switch box assembly. The probe and probe spring shall be disassembled from the switch box assembly. The probe foot shall be disassembled from the shaft and the screw reinserted into the shaft. The cotter key shall be preserved Method IC-1. The probe spring shall be coated with Type P-11 and preserved Method IC-1 of MIL-P-116. The electrical cable shall be coiled around the switch box and securely fastened. The switch box and components shall be placed in a close-fitting fiberboard box conforming to PPP-B-636, W5c. The contents shall be cushioned to prevent free movement and damage.

5.2.1.4.5 Power supply assembly. The openings in the box and cartridge cable end shall be sealed with tape, PPP-T-60, Type IV. The cable shall be coiled around the power supply box and secured with tape specified herein.

5.2.1.4.6 "O" rings. The "O" rings for the piston, piston plug, and barrel shall be preserved in accordance with Method IC-1 of MIL-P-116.

5.2.1.4.7 Faking board. The faking board shall not require preservation.

5.2.1.4.8 Miscellaneous parts. Small miscellaneous parts of like description such as bolts, nuts, washers, grommets, etc., shall be preserved together in accordance with Method IC-1 of MIL-P-116.

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5.2.1.4.9 Pendant. The pendant shall be coiled to the minimum diameter and secured at a minimum of three places equal distantly spaced around the circumference of the coil.

5.2.1.4.10 Consolidation. The preserved piston plug, switch box assembly, power supply assembly, preserved miscellaneous parts, and any other parts requiring consolidation shall be placed in a box conforming to PPP-B-621, Class 2, Style 2, Type 3 load. The contents shall be blocked and cushioned to prevent movement and damage.

5.2.1.5 Anchor test and capacitor charging kit (Type II anchor). The test and capacitor charging kit shall be preserved in accordance with Method IIe of MIL-P-116. The case furnished with the kit shall be the inner container.

5.2.1.6 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, Method IC-1.

5.2.2 Commercial. The anchor, retrofit kit, and anchor test and capacitor charging kit shall be preserved to afford adequate protection against deterioration and damage from the contractor to the initial destination.

5.3 Packing. Packing shall be Level A or Commercial as specified (see 6.2).

5.3.1 Level A.

5.3.1.1 Type I and Type II anchor. The preserved Type I and Type II propellant embedded anchor, electrical wire, and consolidated box shall be packed in a wooden crate meeting the requirements of MIL-C-104, Type I, Class 2, Style a. Dimensions and method of attachment are given in Table I and Figures 1 and 2. The breech block shall ne tightened so that the breech block "O" ring is seated against the barrel face. If the breech is not positioned as depicted in the figures, it shall be backed off until this position is achieved. The joint between the breech block and barrel shall be sealed by tape conforming to PPP-T-60, Type IV.

5.3.1.2 Type II retrofit kit. The Type II retrofit kit shall be packed in a crate conforming to MIL-C-104, Type I, Class 2, Style a.

5.3.1.3 Anchor test and capacitor charging kit (Type II anchor). The preserved test and capacitor charging kit shall be packed in a fiberboard box conforming to PPP-B-636, V3s and closed Method V.

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5.3.1.4 Propellant. When furnished (see 3.7.2), the propellant shall be packed in accordance with MIL-P-323.

5.3.1.5 Blasting caps and detonating fuses. The blasting caps and detonating fuses, when furnished (see 3.7.2), shall be packed in accordance with the applicable provisions of Title 49 of the Code of Federal regulations.

5.3.2 Commercial.

5.3.2.1 Type I and Type II anchor. The Type I and Type II anchor, retrofit kit and the anchor test and capacitor charging kit shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules. The Type I and II anchor shall be packed completely assembled, except for the cartridge.

5.3.2.2 Propellant, blasting caps, and detonating fuses. When furnished (see 3.7.2), propellant, blasting caps, and detonating fuses shall be packed as specified for Level A.

5.4 Marking.

5.4.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129 and Title 49 (Parts 100-199) of the Code of Federal Regulations.

5.4.2 Commercial. Marking for commercial packaging shall be in accordance with FED. STD. No. 356.

6. NOTES

6.1 Intended use.

(a) The Type I propellant embedded anchor is intended for use as a part of the submarine pipeline system to secure the seaward end of the pipeline.
(b) The Type II propellant embedded anchor is intended for use as the main anchor for the multi-leg tanker mooring system.
(c) The retrofit kit for the Type II anchor is intended for use to replace expendable components used in the initial multi-leg mooring system to allow installation of additional anchors, as required.

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(d) The anchor test and capacitor charging kit for the Type II anchor is intended for use to provide a means of testing electronic components of the anchor and for capacitor charging during storage.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of anchor required (see 1.2).
(c) Time frame required for submission of preproduction model (see 3.2).
(d) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(e) When propellant material, blasting caps, detonating fuses, retrofit kits, and anchor test and capacitor charging kits are required (see 3.7.2 and 3.7.3).
(f) Time frame required for submission of preproduction pack (see 5.1).
(g) Degree of preservation and packing required (see 5.2 and 5.3).

6.3 Preproduction model. Any changes or deviations of production anchor, retrofit kit, and anchor test and capacitor charging kit from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish anchors conforming to this specification.

6.4 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the anchors, retrofit kit, and anchor test and capacitor charging kit in accordance with this specification.

6.5 Contract data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each anchor, retrofit kit, and anchor test and capacitor charging kit.

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6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

Custodians:
Army - ME
Navy - YD

Preparing activity:
Army - ME

Project 3835-0047

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TABLE I.
PACKAGING AND PACKING MATERIAL

FIND NO	IDENTIFICATION	DESCRIPTION	QTY
1	MIL-C-104, TYPE I, CLASS 2, STYLE 6	CRATE	1
2	PPP-B-636, TYPE CF, GRADE WSC	BOX	1
3	QQ-S-781, TYPE II, CLASS B	STRAPPING, .018 X $\frac{5}{8}$ WITH NAIL HOLE	AR
4	FF-N-105, TYPE II, STYLE 7	NAILS	AR
5	FF-S-85	SCREW, CAP, HEX, $\frac{3}{8}$-16UNC-2A X 8 LG	4
6	MS 15795-813	WASHER, $\frac{3}{8}$	4
7	MS 35338-46	WASHER, LOCK, $\frac{3}{8}$	4
8	MS 51967-8	NUT, HEX, $\frac{3}{8}$-16 UNC-2B	4
9	MS 35338-50	WASHER, LOCK, $\frac{5}{8}$	1
10	MS 51967-20	NUT, HEX, $\frac{5}{8}$ 11 UNC-2B	2
11	STEEL	ROD, 5/8 DIA X 44 LG (THD BOTH ENDS 5/8-11 UNC-2A X 2)	1
12	STEEL	PIPE, STD, 1 DIA X 28 LG	1
13	STEEL	PLATE, WASHER, $\frac{3}{8}$ X 3 X 6	2
14	STEEL	PLATE, ANCHOR, $\frac{1}{4}$ X 5 X 5	1
15	STEEL	PLATE, WASHER, $\frac{1}{4}$ X 3 X 3	1
16	STEEL	PLATE, DRAG, $\frac{1}{4}$ X 3 X 15	1
17	STEEL	PLATE, BULLET, $\frac{3}{8}$ X 2 X 22	1
18	STEEL	ANGLE, $\frac{1}{4}$ X 2 X 2 X 9$\frac{1}{2}$ LG	2
19	STEEL	ANGLE, $\frac{1}{4}$ X 2 X 2 X 4 LG	1
20	DOUBLE HEADER (SEE MIL-C-104)	LUMBER, SOFTWOOD, 4 X 4 X 54	1
21	FILLER, SHAPED	LUMBER, SOFTWOOD	2
22	FILLER, SHAPED	LUMBER, SOFTWOOD	2
23	FILLER, SHAPED	LUMBER, SOFTWOOD	2
24	FILLER, SHAPED	LUMBER, SOFTWOOD	1
25	FILLER	LUMBER, SOFTWOOD, 2 X 4 X 19	1
26	FILLER	LUMBER, SOFTWOOD, 2 X 4 X 7	1
27	FILLER	LUMBER, SOFTWOOD, 2 X 4 X 13	1
28	FILLER	LUMBER, SOFTWOOD, 2 X 4 X 20	1
29	FILLER	LUMBER, SOFTWOOD, 2 X 4 X 52	1
30	MS 35494-98	SCREW, WOOD, FLAT HEAD, #12 X $\frac{3}{4}$	4
31	FILLER	LUMBER, SOFTWOOD, 2 X 6 X 20	1
32	FILLER	LUMBER, SOFTWOOD, 2 X 6 X 32	2
33	FILLER	LUMBER, SOFTWOOD, 4 X 4 X 23	1
34	FILLER	LUMBER, SOFTWOOD, 4 X 4, CUT TO FIT	AR
35	PPP-B-636, CLASS-WEATHER RESISTANT	BOX	1

X-2443

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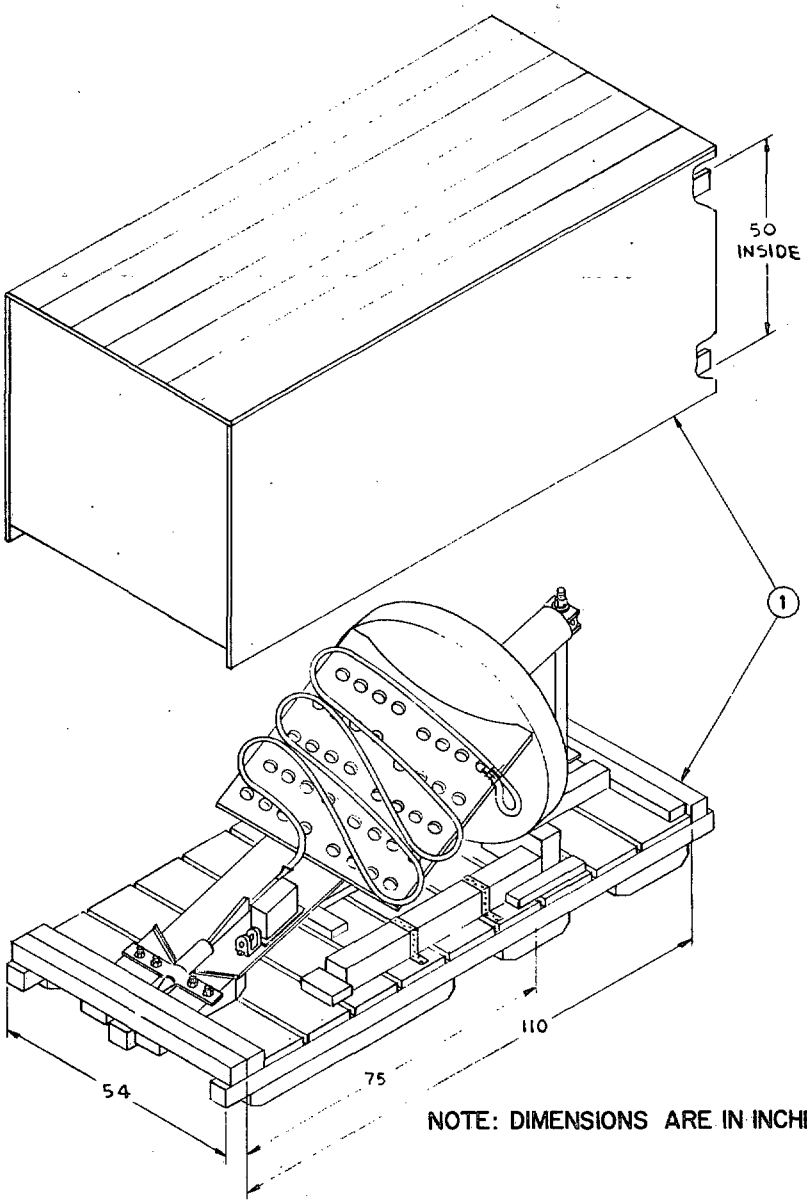


FIGURE 1. Crate dimensions.

X-2444A

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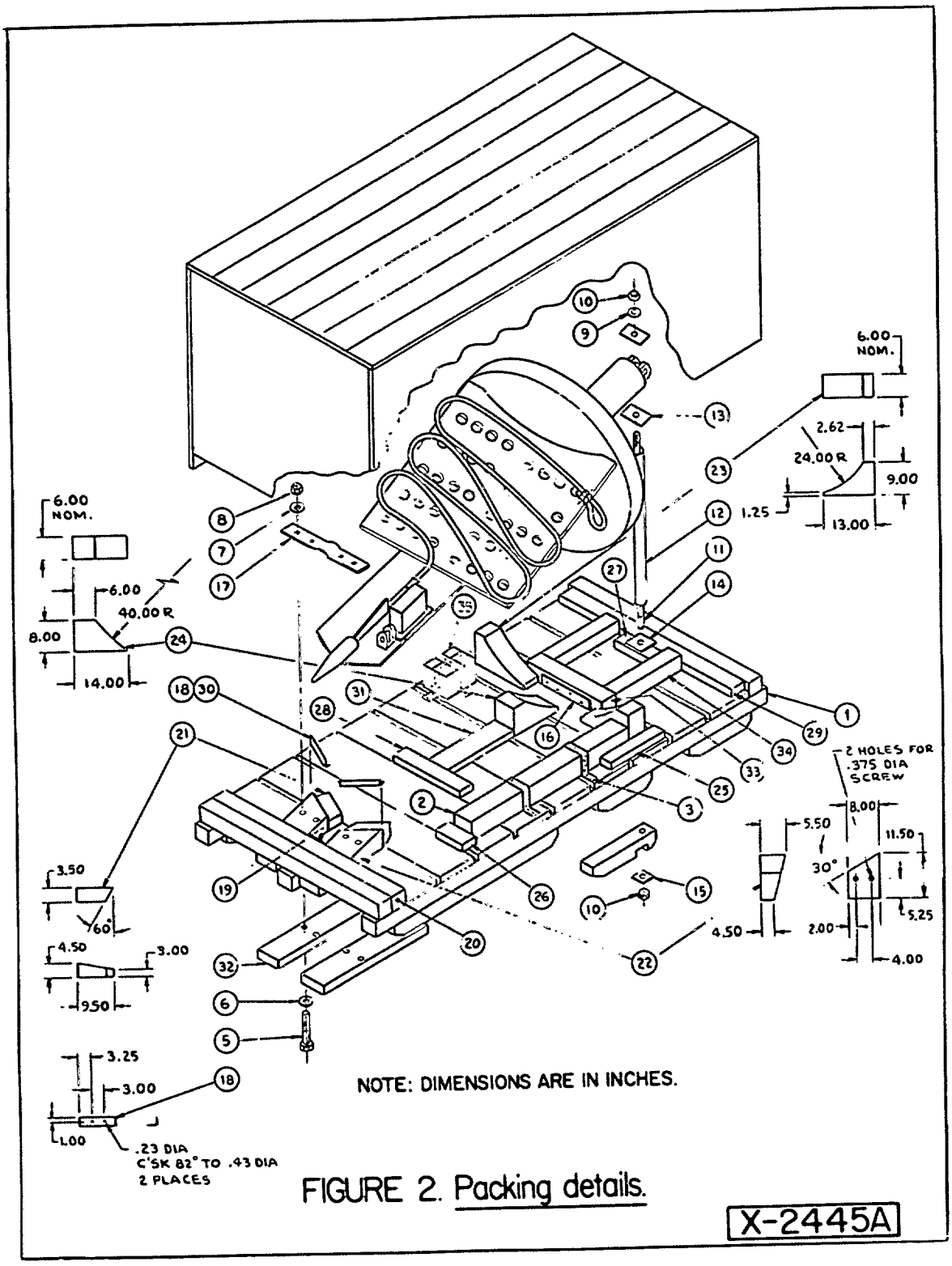


FIGURE 2. Packing details.

X-2445A

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DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-A-52628P Anchor, Propellant Embedment

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

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